

dual-channel drift monitor · the auditor of the auditor

operating cycle: monitor → alarm → bounded refit → epoch reset → monitoring continues

V · VALIDITY · watches the outputs

flag e-values compound into a running wealth · supermartingale under the null
Ville caps false alarms at α under any stopping rule · line 20

the frozen record

real deployment · CN 56 + US 59 windows **0 crossings**

null calibration · 60 runs × 300 windows **0 crossings**

healthy bootstrap · 60 runs × 300 windows **0 crossings**

peak realized wealth · Ville line 20 **0.97**

115 real windows · 120 calibration runs · 0 false alarms

C · CERTIFICATION POWER · watches the auditor

certification strength feeds an e-detector · CUSUM lineage · additive restart
Markov bound over each monitoring epoch · budget 1200

CN stream · 2015-06 crash · one operating cycle

1 regime change · volatility structure rebuilt · never reverted

2 certification strength collapses · median 25.1 → 1.4

3 alarm · window 84 · detector 1207 over budget 1200

4 bounded refit 164 s · strength recovers to 16–27 in 4 windows

5 second alarm · window 94 · epoch resets · watch continues

US stream · COVID transient · detector peak 41
29× below budget · zero alarms · transient absorbed

complementary by construction · V bounds the outputs, C watches the auditor

orbit-member forgeries and value-level corruption are exchangeable-silent on V · both collapse certification power · C closes the gap

injected corruption, windows to alarm: N1 3 · N5 3 · N2 5 · cross-market splice 27